FDN1 — FDN1 TASK 1: DATA PROCESSING SOLUTION

DATA PROCESSING — D608
PRFA — FDN1

Preparation Task Overview Submissions Evaluation Report

COMPETENCIES

4168.1.1 : Recommends a Data Flow and Process Solution

The learner recommends a data processing solution that includes a data flow diagram and a process flow diagram.

4168.1.2 : Implements an ETL Process

The learner implements a process that extracts, transforms, and loads data for use.

4168.1.3 : Creates a Plan for Continuous Monitor

The learner creates a plan for continuous monitoring of data processing pipelines.

INTRODUCTION

Throughout your career in data analytics, you will design data flow and process solutions according to business and data analytic needs. You will demonstrate your ability to design an effective solution that applies the knowledge you have gained about data collection, structuring, and the principles of data flow in cloud-based analytics.

In this task, you will review a set of business requirements for data flow and processing and submit recommendations for a comprehensive data flow solution that optimally integrates with existing systems and processes. Learners will need to use a tool such as Microsoft Visio or Lucidchart to complete this task.

SCENARIO

Refer to the scenario and information in the attached "Data Engineering Scenario" to inform your recommended data architecture design.

Your submission will be a design document submitted in EMA.

REQUIREMENTS

Your submission must represent your original work and understanding of the course material. Most performance assessment submissions are automatically scanned through the WGU similarity checker. Students are strongly encouraged to wait for the similarity report to generate after uploading their work and

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then review it to ensure Academic Authenticity guidelines are met before submitting the file for evaluation. See Understanding Similarity Reports for more information.

Grammarly Note:
Professional Communication will be automatically assessed through Grammarly for Education in most performance assessments before a student submits work for evaluation. Students are strongly encouraged to review the Grammarly for Education feedback prior to submitting work for evaluation, as the overall submission will not pass without this aspect passing. See Use Grammarly for Education Effectively for more information.

Microsoft Files Note:
Write your paper in Microsoft Word (.doc or .docx) unless another Microsoft product, or pdf, is specified in the task directions. Tasks may not be submitted as cloud links, such as links to Google Docs, Google Slides, OneDrive, etc. All supporting documentation, such as screenshots and proof of experience, should be collected in a pdf file and submitted separately from the main file. For more information, please see Computer System and Technology Requirements.

You must use the rubric to direct the creation of your submission because it provides detailed criteria that will be used to evaluate your work. Each requirement below may be evaluated by more than one rubric aspect. The rubric aspect titles may contain hyperlinks to relevant portions of the course.

Note: Refer to the "Data Engineering Scenario" provided in the Supporting Documents section to inform your work.

Note: Written responses must be submitted through EMA.

Part I: Business Case Analysis

A. Provide an overview of the business case by doing the following:
   1. Provide a problem statement based on the attached "Data Engineering Scenario" document indicating the business requirements to be addressed by the new data engineering design.
   2. Create source-to-target data mapping by doing the following:
      a. Define the attributes.
      b. Map the attributes.
      c. Describe necessary data transformations.

Part II: Recommended Design Solution

B. Outline a data engineering design that meets the business needs by doing the following:
   1. Provide a process flow diagram based on the scenario.
   2. Provide a Level 1 or higher data flow diagram showing the logical organization and movement of data for the proposed design as described in part B1.

Part III: Processing Design Evaluation

C. Define the design's relevance to stakeholders by doing the following:

1. Describe **two** or more advantages to the specified approach in your recommended design compared to others you considered that would satisfy the business requirements.
2. Describe **two** or more disadvantages to the specified approach in your recommended design compared to others you considered that would satisfy the business requirements.

Part IV: Submission

D. Submit your findings and recommendations as a file that includes responses to task prompts.

Sources

E. Acknowledge sources, using in-text citations and references, for content that is quoted, paraphrased, or summarized.

Professional Communication

F. Demonstrate professional communication in the content and presentation of your submission.

File Restrictions

File name may contain only letters, numbers, spaces, and these symbols: ! - _ . * ' ( )
File size limit: 200 MB
File types allowed: doc, docx, rtf, xls, xlsx, ppt, pptx, odt, pdf, csv, txt, qt, mov, mpg, avi, mp3, wav, mp4, wma, flv, asf, mpeg, wmv, m4v, svg, tif, tiff, jpeg, jpg, gif, png, zip, rar, tar, 7z

RUBRIC

A1:PROBLEM STATEMENT

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
A problem statement is not provided.	The submission provides a problem statement, but the statement is not relevant to the business requirements represented in the attached "Data Engineering Scenario" or the statement is incomplete or inaccurate.	The submission provides a problem statement that is relevant to business requirements represented in the attached "Data Engineering Scenario."

A2A:DEFINE ATTRIBUTES

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
The data attributes are not defined.	The submission defines the data attributes incompletely or inaccurately.	The submission defines the data attributes completely and accurately.

A2B:MAP ATTRIBUTES

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
The data attributes are not mapped.	The submission maps the data attributes incompletely or inaccurately.	The submission maps the data attributes completely and accurately.

A2C:DATA TRANSFORMATIONS

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
Data transformations are not described.	The submission describes data transformations that are unnecessary, incomplete, or inaccurate.	The submission describes necessary data transformations completely and accurately.

B1:PROCESS FLOW DIAGRAM

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
A process flow diagram is not provided.	The submission provides a process flow diagram that is illogical, inaccurate, or incomplete.	The submission provides a process flow diagram that is logical, accurate, and complete.

B2:DATA FLOW DIAGRAM

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
A data flow diagram is not provided.	The submission provides a data flow diagram that is not Level 1 or higher, or the data flow diagram is illogical, inaccurate, or incomplete.	The submission provides a Level 1 or higher data flow diagram that is logical, accurate, and complete.

C1:DESIGN ADVANTAGES

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT

A description of advantages is not provided.	The submission describes only 1 advantage of the design solution. Or the advantages described are illogical, inaccurate, incomplete, or not applicable to the design solution.	The submission describes at least 2 advantages of the design solution that are logical, accurate, complete, and applicable to the design solution.

C2:DESIGN DISADVANTAGES

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
A description of disadvantages is not provided.	The submission describes disadvantages of the design solution, but 1 or more of the disadvantages provided are illogical, inaccurate, incomplete, or not applicable to the design solution.	The submission describes at least 2 disadvantages of the design solution, and the discussion is logical, accurate, complete, and applicable to the design solution.

D:SUBMISSION

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
No file is provided.	The submission provides a document that is incomplete, is inaccurate, or does not include responses to all task prompts.	The submission provides a file that is complete, is accurate, and includes responses to all task prompts.

E:SOURCES

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
The submission does not include both in-text citations and a reference list for sources that are quoted, paraphrased, or summarized.	The submission includes in-text citations for sources that are quoted, paraphrased, or summarized and a reference list; however, the citations or reference list is incomplete or inaccurate.	The submission includes in-text citations for sources that are properly quoted, paraphrased, or summarized and a reference list that accurately identifies the author, date, title, and source location as available, or the submission states no sources were used.

F:PROFESSIONAL COMMUNICATION

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT

This submission includes pervasive errors in professional communication related to grammar, sentence fluency, contextual spelling, or punctuation, negatively impacting the professional quality and clarity of the writing. Specific errors have been identified by Grammarly for Education under the Correctness category.

This submission includes substantial errors in professional communication related to grammar, sentence fluency, contextual spelling, or punctuation. Specific errors have been identified by Grammarly for Education under the Correctness category.

This submission includes satisfactory use of grammar, sentence fluency, contextual spelling, and punctuation, which promote accurate interpretation and understanding.

SUPPORTING DOCUMENTS

Data Engineering Scenario.docx